

A Partition ℓ_1 Subcase of the L_1/X Daugavet Question

Candidate partial result for arXiv:math/0210287

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Status

This packet gives a candidate partial positive answer to Question (2) in Kadets–Kalton–Werner, *Remarks on rich subspaces of Banach spaces*, arXiv:math/0210287. It is not a full solution of the question.

Source Question

The source paper ends with two questions of A. Pełczyński. The second is:

If X is a subspace of L_1 with the RNP, does L_1/X have the Daugavet property?

The source records positive answers for reflexive spaces, for H^1 , and for a certain Talagrand space.

5. QUESTIONS

We finally mention two questions that were raised by A. Pełczyński which we have not been able to solve.

(1) Is there a rich subspace of L_1 with the Schur property? It was recently proved in [14] that the subspace $X \subset L_1$ constructed by Bourgain and Rosenthal in [5], which has the Schur property and fails the RNP, is a space with the Daugavet property; however, it is not rich in L_1 .

(2) If X is a subspace of L_1 with the RNP, does L_1/X have the Daugavet property? The answer is positive for reflexive spaces [12], for H^1 [22] and a certain space constructed by Talagrand [20] in his (negative) solution of the three-space problem for L_1 [12].

Result

Theorem 1. *Let $(\Omega, \Sigma, \lambda)$ be a nonatomic probability space and let $(A_n)_{n \geq 1}$ be a countable measurable partition of Ω with $\lambda(A_n) > 0$ for every n . Put*

$$e_n = \frac{\mathbf{1}_{A_n}}{\lambda(A_n)} \quad \text{and} \quad X = \overline{\text{span}}\{e_n : n \geq 1\} \subset L_1(\lambda).$$

Then X is isometric to ℓ_1 , hence has the Schur property and the Radon–Nikodym property, and the quotient $L_1(\lambda)/X$ has the Daugavet property.

Proof Intuition

The partition splits $L_1(\lambda)$ as an ℓ_1 -sum of the pieces $L_1(A_n)$. The subspace X is exactly the corresponding ℓ_1 -sum of the one-dimensional constant subspaces on those pieces. Therefore the quotient splits isometrically into an ℓ_1 -sum of the quotients $L_1(A_n)/\text{span}\{\mathbf{1}_{A_n}\}$. Each coordinate quotient is covered by the known finite-dimensional, hence reflexive, subspace case. The only remaining point is that ℓ_1 -sums preserve the Daugavet property, which follows directly from the slice characterization.

Proof

First, the map $(a_n) \mapsto \sum_n a_n e_n$ is an isometry from ℓ_1 onto X , because the A_n 's are disjoint and $\|e_n\|_1 = 1$.

Write $E_n = L_1(A_n, \lambda|_{A_n})$ and $F_n = \text{span}\{\mathbf{1}_{A_n}\}$. Then

$$L_1(\lambda) \cong \left(\bigoplus_{n=1}^{\infty} E_n \right)_{\ell_1}, \quad X \cong \left(\bigoplus_{n=1}^{\infty} F_n \right)_{\ell_1}.$$

The quotient is isometric to

$$\left(\bigoplus_{n=1}^{\infty} E_n/F_n \right)_{\ell_1}.$$

Indeed, for $f = (f_n) \in (\oplus E_n)_{\ell_1}$,

$$\text{dist}(f, X) = \inf_{u_n \in F_n} \sum_n \|f_n - u_n\|_1 = \sum_n \text{dist}(f_n, F_n).$$

The lower bound is immediate. For the upper bound, choose $u_n \in F_n$ within $2^{-n}\eta$ of the coordinate distance and also with $\|u_n\| \leq 2\|f_n\| + 2^{-n}\eta$; this is possible because $0 \in F_n$ is already within $\|f_n\|$ of f_n . Hence $(u_n) \in (\oplus F_n)_{\ell_1}$, and letting $\eta \downarrow 0$ gives equality.

For each n , F_n is one-dimensional and hence reflexive. The positive reflexive-subspace case cited in Kadets–Kalton–Werner therefore gives that E_n/F_n has the Daugavet property.

It remains to recall that an ℓ_1 -sum of Daugavet spaces has the Daugavet property. Let $Y = (\oplus_n Y_n)_{\ell_1}$, where each Y_n has the Daugavet property. Take $y = (y_n) \in S_Y$, a slice

$$S = \{z \in B_Y : \text{Re } y^*(z) > 1 - \alpha\}, \quad y^* = (y_n^*) \in S_{Y^*} = (\oplus_n Y_n^*)_{\ell_\infty},$$

and $\varepsilon > 0$. Choose k with $\|y_k^*\| > 1 - \alpha/2$. By the homogeneous form of the Daugavet slice characterization in Y_k , applied to $y_k \in Y_k$ and to the slice determined by $y_k^*/\|y_k^*\|$, there is $z_k \in S_{Y_k}$ such that

$$\text{Re } \frac{y_k^*}{\|y_k^*\|}(z_k) > 1 - \frac{\alpha}{2\|y_k^*\|} \quad \text{and} \quad \|y_k + z_k\| > \|y_k\| + 1 - \varepsilon.$$

Then $\text{Re } y_k^*(z_k) > 1 - \alpha$. Let $z \in Y$ be supported only in coordinate k , with z_k as above. Then $z \in S$ and

$$\|y + z\| = \sum_{n \neq k} \|y_n\| + \|y_k + z_k\| > 1 - \|y_k\| + \|y_k\| + 1 - \varepsilon = 2 - \varepsilon.$$

Thus Y has the Daugavet property. Applying this to $Y_n = E_n/F_n$ proves that $L_1(\lambda)/X$ has the Daugavet property.

Verification Notes

The proof uses only standard quotient-norm computations and the slice characterization of the Daugavet property. The subspace X is a nonreflexive RNP example, since it is isometric to ℓ_1 . This is a genuine subcase of the source question, but it is far from the general RNP-subspace problem.

Novelty and Literature Check

The cheap run indexes had no exact packet for arXiv:math/0210287. Web searches on 2026-06-16 for phrases including “rich subspace of L_1 with the Schur property”, “ L_1/X Daugavet property RNP”, and “quotient L_1 Daugavet RNP subspace” found the source paper and adjacent Daugavet/rich-subspace literature, but no exact closure of the full question and no exact match for the partition ℓ_1 subcase. This packet should be reviewed as a small positive partial result, not as a claimed new full theorem.

Limitations

The theorem only handles the very structured Schur/RNP subspaces generated by a countable measurable partition. It does not answer whether every RNP subspace $X \subset L_1$ has L_1/X with the Daugavet property, and it does not address the separate question of whether there is a rich subspace of L_1 with the Schur property.

References

References

- [1] V. Kadets, N. Kalton, and D. Werner, *Remarks on rich subspaces of Banach spaces*, arXiv:math/0210287.
- [2] V. M. Kadets, R. V. Shvidkoy, G. G. Sirotkin, and D. Werner, *Banach spaces with the Daugavet property*, Trans. Amer. Math. Soc. 352 (2000), 855–873.